

2.2 Calculus of Vector-Valued Functions

1. Determine if the following limits exist. If it does not exist, you must explain why.

a) If $\mathbf{r}(t) = \frac{t-2}{t^2-4}\mathbf{i} + \frac{\cos(\pi/t)}{t-2}\mathbf{j} - \tan^{-1}\left(\frac{1}{|t-2|}\right)\mathbf{k}$, then find $\lim_{t \rightarrow 2} \mathbf{r}(t)$.

We calculate that:

$$\begin{aligned}\lim_{t \rightarrow 2} \mathbf{r}(t) &= \left\langle \lim_{t \rightarrow 2} \frac{t-2}{t^2-4}, \lim_{t \rightarrow 2} \frac{\cos(\pi/t)}{t-2}, -\lim_{t \rightarrow 2} \tan^{-1}\left(\frac{1}{|t-2|}\right) \right\rangle \\ &\stackrel{H}{=} \left\langle \lim_{t \rightarrow 2} \frac{1}{t+2}, \lim_{t \rightarrow 2} \frac{-\sin(\pi/t)(-\pi/t^2)}{1}, -\lim_{t \rightarrow \infty} \tan^{-1}(k) \right\rangle \\ &= \left\langle \frac{1}{4}, -\sin(\pi/2)(-\pi/4), -\frac{\pi}{2} \right\rangle \\ &= \left\langle \frac{1}{4}, \frac{\pi}{4}, -\frac{\pi}{2} \right\rangle\end{aligned}$$

b) If $\mathbf{r}(t) = e^{-3t}\mathbf{i} + \frac{t^2}{\sin^2(t)}\mathbf{j} + \cos(2t)\mathbf{k}$, then find $\lim_{t \rightarrow 0} \mathbf{r}(t)$.

We calculate that:

$$\begin{aligned}\lim_{t \rightarrow 0} \mathbf{r}(t) &= \left\langle \lim_{t \rightarrow 0} e^{-3t}, \lim_{t \rightarrow 0} \frac{t^2}{\sin^2(t)}, \lim_{t \rightarrow 0} \cos(2t) \right\rangle \\ &\stackrel{H}{=} \left\langle e^0, \lim_{t \rightarrow 0} \frac{2t}{2\sin(t)\cos(t)}, \cos(0) \right\rangle \\ &\stackrel{H}{=} \left\langle 1, \lim_{t \rightarrow 0} \frac{2}{2\cos^2(t) - 2\sin^2(t)}, 1 \right\rangle \\ &= \langle 1, 1, 1 \rangle\end{aligned}$$

c) Find $\lim_{t \rightarrow \infty} \left\langle \frac{1+t^2}{1-t^2}, \tan^{-1}(t), \frac{1-e^{-2t}}{t} \right\rangle$.

We calculate that:

$$\begin{aligned}\lim_{t \rightarrow \infty} \left\langle \frac{1+t^2}{1-t^2}, \tan^{-1}(t), \frac{1-e^{-2t}}{t} \right\rangle &= \left\langle \lim_{t \rightarrow \infty} \frac{1+t^2}{1-t^2}, \lim_{t \rightarrow \infty} \tan^{-1}(t), \lim_{t \rightarrow \infty} \frac{1-e^{-2t}}{t} \right\rangle \\ &= \left\langle \lim_{t \rightarrow \infty} \frac{1+t^2}{1-t^2} \cdot \frac{1/t^2}{1/t^2}, \frac{\pi}{2}, \lim_{t \rightarrow \infty} \frac{1-0}{t} \right\rangle \\ &= \left\langle -1, \frac{\pi}{2}, 0 \right\rangle\end{aligned}$$

d) Find $\lim_{t \rightarrow \infty} \left\langle \frac{\ln(t)}{2\sqrt{t}}, te^{-t}, \frac{2t^5 - 5t}{1+t^5} \right\rangle$.

We calculate that:

$$\begin{aligned} \lim_{t \rightarrow \infty} \left\langle \frac{\ln(t)}{2\sqrt{t}}, te^{-t}, \frac{2t^5 - 5t}{1+t^5} \right\rangle &= \left\langle \lim_{t \rightarrow \infty} \frac{\ln(t)}{2\sqrt{t}}, \lim_{t \rightarrow \infty} \frac{t}{e^t}, \lim_{t \rightarrow \infty} \frac{2t^5 - 5t}{1+t^5} \right\rangle \\ &\stackrel{H}{=} \left\langle \lim_{t \rightarrow \infty} \frac{\left(\frac{1}{t}\right)}{\left(\frac{1}{\sqrt{t}}\right)}, \lim_{t \rightarrow \infty} \frac{t}{e^t}, \lim_{t \rightarrow \infty} \frac{2t^5 - 5t}{1+t^5} \cdot \frac{1/t^5}{1/t^5} \right\rangle \\ &= \left\langle \lim_{t \rightarrow \infty} \left(\frac{\sqrt{t}}{t}\right), \lim_{t \rightarrow \infty} \frac{1}{e^t}, \lim_{t \rightarrow \infty} \frac{2 - 5/t^4}{1/t^5 + 1} \right\rangle \\ &= \left\langle \lim_{t \rightarrow \infty} \frac{1}{\sqrt{t}}, 0, 2 \right\rangle \\ &= \langle 0, 0, 2 \rangle \end{aligned}$$

2. Find the parametric equation for the tangent line to the curve with given parametric equations at the specified point.

$$x = 2 + \ln(t), \quad y = 3t^3, \quad z = 3t + 1; \quad (2, 3, 4)$$

We see that on the given curve we arrive at the point $(2, 3, 4)$ when:

$$\begin{aligned} 2 &= 2 + \ln(t) & 0 &= \ln(t) \\ 3 &= 3t^3 & \Rightarrow & 1 = t^3 & \Rightarrow & t = 1 \\ 4 &= 3t + 1 & 3 &= 3t \end{aligned}$$

The tangent vector to the at this point in time is given by evaluating $x' = \frac{1}{t}$, $y' = 9t^2$, & $z' = 3$ evaluated at $t = 1$. This produces a direction vector of $v = \langle 1, 9, 3 \rangle$.

Therefore the equation of the tangent line to the curve at $(2, 3, 4)$ is $L = \begin{cases} x = t + 2 \\ y = 9t + 3 \\ z = 3t + 4 \end{cases}$.

3. Let $\mathbf{r}(t) = \langle 5 \tan^{-1}(t-1), \sinh(t-2), -\ln(t-1) \rangle$. Write parametric equations of the tangent line to the graph of $\mathbf{r}(t)$ at $\left(\frac{5\pi}{4}, 0, 0\right)$.

We realize that on the given curve we arrive at the point $\left(\frac{5\pi}{4}, 0, 0\right)$ when:

$$\begin{aligned} \frac{5\pi}{4} &= 5 \tan^{-1}(t-1) & \frac{\pi}{4} &= \tan^{-1}(t-1) \\ 0 &= \sinh(t-2) & \Rightarrow 0 &= \sinh(t-1) \Rightarrow t = 2 \\ 0 &= -\ln(t-1) & 0 &= \ln(t-1) \end{aligned}$$

We calculate that $\mathbf{r}'(t) = \left\langle \frac{5}{1+(t-1)^2}, \cosh(t-2), -\frac{1}{t-1} \right\rangle$.

Therefore, the tangent vector at $t = 2$ is given by $\mathbf{r}'(2) = \left\langle \frac{5}{2}, 1, -1 \right\rangle$.

Therefore the equation of the tangent line to the curve at $\left(\frac{5\pi}{4}, 0, 0\right)$ is $L = \begin{cases} x = \frac{5}{2}t + \frac{5\pi}{4} \\ y = t \\ z = -t \end{cases}$

4. Find a vector function $\mathbf{L}(t)$ which is the tangent line to the following curves at the following points.

- a) $\mathbf{r}(t) = \langle 1 + 2\sqrt{t}, t^3 - t, t^3 + t \rangle$ at the point $(3, 0, 2)$

We note that $\mathbf{r}(t)$ arrives at the point $(3, 0, 2)$ when $t = 1$.

We calculate that $\mathbf{r}'(t) = \left\langle \frac{1}{\sqrt{t}}, 3t^2 - 1, 3t^2 + 1 \right\rangle$.

Therefore, the tangent vector at $t = 1$ is given by $\mathbf{r}'(1) = \langle 1, 2, 4 \rangle$.

Therefore, the equation of the tangent line to the curve at $(3, 0, 2)$ is $\mathbf{L}(t) = \langle t + 3, 2t, 4t + 2 \rangle$.

- b) $\mathbf{r}(t) = \langle e^t, te^t, te^{t^2} \rangle$ at the point $(1, 0, 0)$

We note that $\mathbf{r}(t)$ arrives at the point $(1, 0, 0)$ when $t = 0$.

We calculate that $\mathbf{r}'(t) = \langle e^t, e^t + te^t, e^{t^2} + t(2t)e^t \rangle$.

Therefore, the tangent vector at $t = 0$ is given by $\mathbf{r}'(0) = \langle 1, 1, 1 \rangle$.

Therefore, the equation of the tangent line to the curve at $(1, 0, 0)$ is $\mathbf{L}(t) = \langle t + 1, t, t \rangle$.

c) $\mathbf{r}(t) = \langle e^{-t} \cos(t), e^{-t} \sin(t), e^{-t} \rangle$ at the point $(1, 0, 1)$

We note that $\mathbf{r}(t)$ arrives at the point $(1, 0, 1)$ when $t = 0$.

We calculate that $\mathbf{r}'(t) = \langle -e^{-t} \cos(t) - e^{-t} \sin(t), -e^{-t} \sin(t) + e^{-t} \cos(t), -e^{-t} \rangle$.

Therefore, the tangent vector at $t = 0$ is given by $\mathbf{r}'(0) = \langle -1, 1, -1 \rangle$.

Therefore, the equation of the tangent line to the curve at $(1, 0, 1)$ is $\mathbf{L}(t) = \langle -t + 1, t, -t + 1 \rangle$.

d) $\mathbf{r}(t) = \langle \sqrt{t^2 + 3}, \ln(t^2 + 3), t \rangle$ at the point $(2, \ln(4), 1)$

We note that $\mathbf{r}(t)$ arrives at the point $(2, \ln(4), 1)$ when $t = 1$.

We calculate that $\mathbf{r}'(t) = \langle \frac{t}{\sqrt{t^2 + 3}}, \frac{2t}{t^2 + 3}, 1 \rangle$.

Therefore, the tangent vector at $t = 1$ is given by $\mathbf{r}'(1) = \langle \frac{1}{2}, \frac{1}{2}, 1 \rangle$.

Therefore, the equation of the tangent line to the curve at $(2, \ln(4), 1)$ is

$$\mathbf{L}(t) = \left\langle \frac{1}{2}t + 2, \frac{1}{2}t + \ln(4), t + 1 \right\rangle.$$

5. Let $\mathbf{r}(t) = \langle e^{2t} \cos(2t), 5, e^{2t} \sin(2t) \rangle$, find $\|\mathbf{r}'(t)\|$. Fully simplify the final solution.

We see that $\mathbf{r}'(t) = \langle 2e^{2t} \cos(2t) - 2e^{2t} \sin(2t), 0, 2e^{2t} \sin(2t) + 2e^{2t} \cos(2t) \rangle$ and therefore,

$$\begin{aligned} \|\mathbf{r}'(t)\| &= \sqrt{[2e^{2t} \cos(2t) - 2e^{2t} \sin(2t)]^2 + [2e^{2t} \sin(2t) + 2e^{2t} \cos(2t)]^2} \\ &= \sqrt{8e^{4t} \cos^2(2t) + 8e^{4t} \sin^2(2t)} \\ &= \sqrt{8e^{4t}} \\ &= 2\sqrt{2}e^{2t} \end{aligned}$$

6. Two particles travel along the space curves

$$\mathbf{r}_1(t) = \langle t, t^2, t^3 \rangle, \quad \mathbf{r}_2(t) = \langle 1+2t, 1+6t, 1+14t \rangle$$

Do the particles collide? Do their paths intersect?

We redefine the 2nd curve here to be in terms of a new variable of s : $\mathbf{r}_2(s) = \langle 1+2s, 1+6s, 1+14s \rangle$

To see if these paths intersect we need to have $\mathbf{r}_1$ & $\mathbf{r}_2$ arrive at the same point in space and therefore have the same x , y , and z coordinates at some time(s).

We must solve the system of equations
$$\begin{cases} t = 1 + 2s \\ t^2 = 1 + 6s \\ t^3 = 1 + 14s \end{cases}$$

Using the first two equations we see that

$$\begin{aligned} (1+2s)^2 &= 1+6s \rightarrow 4s^2 + 4s + 1 = 1+6s \rightarrow 4s^2 - 2s = 0 \rightarrow 2s(2s-1) = 0 \\ &\rightarrow s = 0 \text{ or } s = \frac{1}{2} \end{aligned}$$

We then note from the first equation that if $s = 0$ then $t = 1$ and if $s = \frac{1}{2}$ then $t = 2$.

Neither of these pairs make the 3rd equation true and solve the system.

Therefore, this tells us that the particles never reach the same location in space and therefore never intersect.

7. The curves $\mathbf{r}_1(s) = \langle 3(s-1), 2(s-1), 5(s-1)^2 \rangle$ and $\mathbf{r}_2(t) = \langle \sin(t), \sin(2t), t \rangle$ intersect at the origin.

Find their angle of intersection.

To answer this question, we must determine the angle of intersection of the tangent vectors at the origin.

We see that $\mathbf{r}_1(s) = \mathbf{0}$ when $s = 1$ and $\mathbf{r}_2(t) = \mathbf{0}$ when $t = 0$.

So, the tangent vectors to the curves at the origin are given by:

$$\mathbf{a} = \mathbf{r}_1'(1) = \langle 3, 2, 10(1-1) \rangle = \langle 3, 2, 0 \rangle$$

$$\mathbf{b} = \mathbf{r}_2'(0) = \langle \cos(0), 2\cos(2(0)), 1 \rangle = \langle 1, 2, 1 \rangle$$

We also see that $\|\mathbf{a}\| = \sqrt{13}$ & $\|\mathbf{b}\| = \sqrt{6}$.

Therefore, the angle which exists between these vectors is given by:

$$\theta = \cos^{-1} \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \right) = \cos^{-1} \left(\frac{3+4+0}{\sqrt{13} \cdot \sqrt{6}} \right) = \cos^{-1} \left(\frac{7}{\sqrt{78}} \right)$$

8. Evaluate each of the following integrals.

a) $\int_0^2 (\mathbf{i} - t^3 \mathbf{j} + 3t^5 \mathbf{k}) dt$

$$\int_0^2 (\mathbf{i} - t^3 \mathbf{j} + 3t^5 \mathbf{k}) dt = \left. \frac{1}{2} t^2 \mathbf{i} - \frac{1}{4} t^4 \mathbf{j} + \frac{1}{2} t^6 \mathbf{k} \right|_0^2 = \langle 2, 4, 32 \rangle - \langle 0, 0, 0 \rangle = \langle 2, 4, 32 \rangle$$

b) $\int_0^1 \left(\frac{4}{1+t^2} \mathbf{j} + \frac{2t}{1+t^2} \mathbf{k} \right) dt$

$$\int_0^1 \left(\frac{4}{1+t^2} \mathbf{j} + \frac{2t}{1+t^2} \mathbf{k} \right) dt = \left\langle 0, 4 \tan^{-1}(t), \int_0^1 \frac{2t}{1+t^2} dt \right\rangle$$

We use a u-sub to help evaluate the last integral.

Let $u = 1+t^2$ & $du = 2t dt$. The bounds become $u(0) = 1$ & $u(1) = 2$.

$$\left\langle 0, 4 \tan^{-1}(t), \int_0^1 \frac{2t}{1+t^2} dt \right\rangle = \left\langle 0, 4 \tan^{-1}(t), \int_1^2 \frac{1}{u} du \right\rangle = \left\langle 0, 4 \tan^{-1}(t), \ln \|u\|_1 \right\rangle = \langle 0, 4 \tan^{-1}(t), \ln(2) \rangle.$$

c) $\int_0^{\pi/2} \langle 3 \sin^2(t) \cos(t), 3 \sin(t) \cos^2(t), 2 \sin(t) \cos(t) \rangle dt$

$$\begin{aligned} \int_0^{\pi/2} \langle 3 \sin^2(t) \cos(t), 3 \sin(t) \cos^2(t), 2 \sin(t) \cos(t) \rangle dt \\ = \left\langle \int_0^{\pi/2} 3 \sin^2(t) \cos(t) dt, \int_0^{\pi/2} 3 \sin(t) \cos^2(t) dt, \int_0^{\pi/2} 2 \sin(t) \cos(t) dt \right\rangle \end{aligned}$$

For each integral we use a substitution.

For the first and third integrals we let $u = \sin(t)$ & $du = \cos(t) dt$. The bounds become

$$u(0) = 0 \text{ \& } u\left(\frac{\pi}{2}\right) = 1.$$

For the second integral we let $u = \cos(t)$ & $du = -\sin(t) dt \rightarrow -du = \sin(t) dt$. The bounds become

$$u(0) = 1 \text{ \& } u\left(\frac{\pi}{2}\right) = 0.$$

Therefore,

$$\begin{aligned} \left\langle \int_0^{\pi/2} 3 \sin^2(t) \cos(t) dt, \int_0^{\pi/2} 3 \sin(t) \cos^2(t) dt, \int_0^{\pi/2} 2 \sin(t) \cos(t) dt \right\rangle &= \left\langle \int_0^1 3u^2 du, \int_1^0 -3u^2 du, \int_0^1 2u du \right\rangle \\ &= \left\langle u^3 \Big|_0^1, u^3 \Big|_{-1}^0, u^2 \Big|_0^1 \right\rangle = \langle 1 - 0, 0 - (-1), 1 - 0 \rangle = \langle 1, 1, 1 \rangle \end{aligned}$$

$$\text{d) } \int_1^2 \langle t^2, t\sqrt{t-1}, t \sin(\pi t) \rangle dt$$

$$\int_1^2 \langle t^2, t\sqrt{t-1}, t \sin(\pi t) \rangle dt = \left\langle \int_1^2 t^2 dt, \int_1^2 t\sqrt{t-1} dt, \int_1^2 t \sin(\pi t) dt \right\rangle$$

For the first integral we see that $\int_1^2 t^2 dt = \frac{1}{3} t^3 \Big|_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}$.

For the second integral we proceed with a u-sub. We let $u = t - 1 \rightarrow t = u + 1$ & $du = dt$.

The bounds become $u(1) = 0$ & $u(2) = 1$.

Therefore, $\int_1^2 t\sqrt{t-1} dt = \int_0^1 (u+1)\sqrt{u} du = \int_0^1 (u^{3/2} + u^{1/2}) du = \left(\frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} \right) \Big|_0^1 = \frac{2}{5} + \frac{2}{3} = \frac{16}{15}$.

For the third integral we proceed with Integration by Parts.

We let $u = t$ & $dv = \sin(\pi t) dt$, then $du = dt$ & $v = -\frac{1}{\pi} \cos(\pi t)$

Therefore,

$$\int_1^2 t \sin(\pi t) dt = -\frac{t}{\pi} \cos(\pi t) \Big|_1^2 + \int_1^2 \frac{1}{\pi} \cos(\pi t) dt = \left(-\frac{2}{\pi} - \frac{1}{\pi} \right) + \frac{1}{\pi^2} \sin(\pi t) \Big|_1^2 = -\frac{3}{\pi} + (0 - 0) = -\frac{3}{\pi}$$

This implies that $\int_1^2 \langle t^2, t\sqrt{t-1}, t \sin(\pi t) \rangle dt = \left\langle \frac{7}{3}, \frac{16}{15}, -\frac{3}{\pi} \right\rangle$.

$$\text{e) } \int \left(\sec^2(t) \mathbf{i} + t(t^2+1)^3 \mathbf{j} + t^2 \ln(t) \mathbf{k} \right) dt$$

$$\int \left(\sec^2(t) \mathbf{i} + t(t^2+1)^3 \mathbf{j} + t^2 \ln(t) \mathbf{k} \right) dt = \left\langle \int \sec^2(t) dt, \int t(t^2+1)^3 dt, \int t^2 \ln(t) dt \right\rangle$$

For the first integral we see that $\int \sec^2(t) dt = \tan(t) + C_1$.

For the second integral we proceed with a u-sub. We let $u = t^2 + 1$ & $du = 2t dt \rightarrow \frac{1}{2} du = t dt$.

Therefore, $\int t(t^2+1)^3 dt = \int \frac{1}{2} u^3 du = \frac{1}{8} u^4 + C_2 = \frac{1}{8} (t^2+1)^4 + C_2$.

For the third integral we proceed with Integration by Parts.

We let $u = \ln(t)$ & $dv = t^2 dt$, then $du = \frac{1}{t} dt$ & $v = \frac{1}{3} t^3$

Therefore,

$$\int t^2 \ln(t) dt = \frac{1}{3} t^3 \ln(t) - \int \frac{1}{3} t^3 \left(\frac{1}{t} \right) dt = \frac{1}{3} t^3 \ln(t) - \int \frac{1}{3} t^2 dt = \frac{1}{3} t^3 \ln(t) - \frac{1}{9} t^3 + C_3$$

This implies that

$$\begin{aligned} \int \left(\sec^2(t) \mathbf{i} + t(t^2+1)^3 \mathbf{j} + t^2 \ln(t) \mathbf{k} \right) dt &= \left\langle \tan(t) + C_1, \frac{1}{8} (t^2+1)^4 + C_2, \frac{1}{3} t^3 \ln(t) - \frac{1}{9} t^3 + C_3 \right\rangle \\ &= \left\langle \tan(t), \frac{1}{8} (t^2+1)^4, \frac{1}{3} t^3 \ln(t) - \frac{1}{9} t^3 \right\rangle + \langle C_1, C_2, C_3 \rangle \\ &= \left\langle \tan(t), \frac{1}{8} (t^2+1)^4, \frac{1}{3} t^3 \ln(t) - \frac{1}{9} t^3 \right\rangle + \mathbf{C} \end{aligned}$$

Note: Any of the 3 forms of the answer shown above are acceptable.

$$f) \int \left\langle te^{2t}, \frac{t}{1-t}, \frac{1}{\sqrt{1-t^2}} \right\rangle dt$$

$$\int \left\langle te^{2t}, \frac{t}{1-t}, \frac{1}{\sqrt{1-t^2}} \right\rangle dt = \left\langle \int te^{2t} dt, \int \frac{t}{1-t} dt, \int \frac{1}{\sqrt{1-t^2}} dt \right\rangle$$

For the first integral we proceed with Integration by Parts.

$$\text{We let } u = t \text{ \& } dv = e^{2t} dt, \text{ then } du = dt \text{ \& } v = \frac{1}{2} e^{2t}$$

$$\text{Therefore, } \int te^{2t} dt = \frac{1}{2} te^{2t} - \int \frac{1}{2} e^{2t} dt = \frac{1}{2} te^{2t} - \frac{1}{4} e^{2t} + C_1$$

For the second integral we proceed with a u-sub. We let $u = 1-t$ & $du = -dt \rightarrow -du = dt$. This also implies that $t = 1-u$.

$$\text{Therefore, } \int \frac{t}{1-t} dt = \int -\frac{1-u}{u} du = \int \left(1 - \frac{1}{u}\right) du = u - \ln|u| + C_2 = 1-t - \ln|1-t| + C_2.$$

$$\text{For the third integral we see that } \int \frac{1}{\sqrt{1-t^2}} dt = \sin^{-1}(t) + C_3.$$

This implies that

$$\begin{aligned} \int \left\langle te^{2t}, \frac{t}{1-t}, \frac{1}{\sqrt{1-t^2}} \right\rangle dt &= \left\langle \frac{1}{2} te^{2t} - \frac{1}{4} e^{2t} + C_1, 1-t - \ln|1-t| + C_2, \sin^{-1}(t) + C_3 \right\rangle \\ &= \left\langle \frac{1}{2} te^{2t} - \frac{1}{4} e^{2t}, 1-t - \ln|1-t|, \sin^{-1}(t) \right\rangle + \langle C_1, C_2, C_3 \rangle \\ &= \left\langle \frac{1}{2} te^{2t} - \frac{1}{4} e^{2t}, 1-t - \ln|1-t|, \sin^{-1}(t) \right\rangle + \mathbf{C} \end{aligned}$$

Note: Any of the 3 forms of the answer shown above are acceptable.

$$9. \text{ Let } \mathbf{r}(t) = \left\langle 2, \sqrt{1+t}, \frac{2}{t^2-3t-4} \right\rangle. \text{ Evaluate } \int_0^3 \mathbf{r}(t) dt.$$

$$\text{We know that } \int_0^3 \mathbf{r}(t) dt = \left\langle \int_0^3 2 dt, \int_0^3 \sqrt{1+t} dt, \int_0^3 \frac{2}{t^2-3t-4} dt \right\rangle \text{ so we must evaluate each of the three}$$

integrals. The first two integrals are very straight forward. The third integral here will require us to integrate using partial fractions.

$$\text{Note that: } \frac{2}{t^2-3t-4} = \frac{2}{(t-4)(t+1)} \stackrel{?}{=} \frac{A}{(t-4)} + \frac{B}{(t+1)} = \frac{A(t+1) + B(t-4)}{(t-4)(t+1)}.$$

In order for these to be equal it must be the case that $2 = A(t+1) + B(t-4)$

- If $t = 4$ then $2 = 5A \Rightarrow A = \frac{2}{5}$
- If $t = -1$ then $2 = -5B \Rightarrow B = -\frac{2}{5}$

So we conclude that $\frac{2}{t^2-3t-4} = \frac{2/5}{(t-4)} - \frac{2/5}{(t+1)}$.

Therefore, we calculate:

$$\begin{aligned}\int_0^3 \mathbf{r}(t) dt &= \left\langle \int_0^3 2dt, \int_0^3 \sqrt{1+t} dt, \int_0^3 \frac{2}{t^2-3t-4} dt \right\rangle \\ &= \left\langle 2t \Big|_0^3, \frac{2}{3}(1+t)^{3/2} \Big|_0^3, \int_0^3 \left[\frac{2/5}{t-4} - \frac{2/5}{t+1} \right] dt \right\rangle \\ &= \left\langle 6, \frac{2}{3}(4)^{3/2} - \frac{2}{3}, \frac{2}{5} \ln|t-4| - \frac{2}{5} \ln|t+1| \Big|_0^3 \right\rangle \\ &= \left\langle 6, \frac{14}{3}, \left(\frac{2}{5} \ln|-1| - \frac{2}{5} \ln|4| \right) - \left(\frac{2}{5} \ln|-4| - \frac{2}{5} \ln|1| \right) \right\rangle \\ &= \left\langle 6, \frac{14}{3}, -\frac{4}{5} \ln(4) \right\rangle\end{aligned}$$

10. Determine the function $\mathbf{r}(t)$ which satisfies each of the given conditions.

a) $\mathbf{r}'(t) = \langle 2t, 3t^2, 1 \rangle$ and $\mathbf{r}(0) = \langle 1, -2, 4 \rangle$.

We note that $\mathbf{r}(t) = \langle t^2 + C_1, t^3 + C_2, t + C_3 \rangle$.

Since $\mathbf{r}(0) = \langle 1, -2, 4 \rangle$ this implies that

$$0 + C_1 = 1 \quad \rightarrow \quad C_1 = 1$$

$$0 + C_2 = -2 \quad \rightarrow \quad C_2 = -2$$

$$0 + C_3 = 4 \quad \rightarrow \quad C_3 = 4$$

Therefore, $\mathbf{r}(t) = \langle t^2 + 1, t^3 - 2, t + 4 \rangle$.

b) $\mathbf{r}'(t) = \langle \cos(t), 2\sin(t), \sec^2(t) \rangle$ and $\mathbf{r}(\pi) = \langle 3, 1, -1 \rangle$.

We note that $\mathbf{r}(t) = \langle \sin(t) + C_1, -2\cos(t) + C_2, \tan(t) + C_3 \rangle$.

Since $\mathbf{r}(\pi) = \langle 3, 1, -1 \rangle$ this implies that

$$0 + C_1 = 3 \quad \rightarrow \quad C_1 = 3$$

$$2 + C_2 = 1 \quad \rightarrow \quad C_2 = -1$$

$$0 + C_3 = -1 \quad \rightarrow \quad C_3 = -1$$

Therefore, $\mathbf{r}(t) = \langle \sin(t) + 3, -2\cos(t) - 1, \tan(t) - 1 \rangle$.

c) $\mathbf{r}''(t) = \langle 6t, e^t, -\cos(t) \rangle$ and $\mathbf{r}'(0) = \langle 0, 1, 2 \rangle$ & $\mathbf{r}(0) = \langle 1, 0, -1 \rangle$.

We note that $\mathbf{r}(t) = \langle 3t^2 + C_1, e^t + C_2, -\sin(t) + C_3 \rangle$.

Since $\mathbf{r}'(0) = \langle 0, 1, 2 \rangle$ this implies that

$$0 + C_1 = 0 \rightarrow C_1 = 0$$

$$1 + C_2 = 1 \rightarrow C_2 = 0$$

$$0 + C_3 = 2 \rightarrow C_3 = 2$$

Therefore, $\mathbf{r}'(t) = \langle 3t^2, e^t, -\sin(t) + 2 \rangle$.

This implies that $\mathbf{r}(t) = \langle t^3 + D_1, e^t + D_2, \cos(t) + 2t + D_3 \rangle$.

Since $\mathbf{r}(0) = \langle 1, 0, -1 \rangle$ this implies that

$$0 + D_1 = 1 \rightarrow D_1 = 1$$

$$1 + D_2 = 0 \rightarrow D_2 = -1$$

$$1 + D_3 = -1 \rightarrow D_3 = -2$$

Therefore, $\mathbf{r}'(t) = \langle t^3 + 1, e^t - 1, \cos(t) + 2t + 2 \rangle$.

d) $\mathbf{r}'(t) = \left\langle e^{2t}, \frac{1}{t+1}, te^{t^2} \right\rangle$ and $\mathbf{r}(0) = \langle 0, 2, -3 \rangle$.

We note that $\mathbf{r}(t) = \left\langle \frac{1}{2}e^{2t} + C_1, \ln|t+1| + C_2, \frac{1}{2}e^{t^2} + C_3 \right\rangle$.

On the final component we use a u-sub with $u = t^2$ & $du = 2tdt \rightarrow \frac{1}{2}du = tdt$

Since $\mathbf{r}(0) = \langle 1, 0, -1 \rangle$ this implies that

$$\frac{1}{2} + C_1 = 0 \rightarrow C_1 = -\frac{1}{2}$$

$$0 + C_2 = 2 \rightarrow C_2 = 2$$

$$\frac{1}{2} + C_3 = -3 \rightarrow C_3 = -\frac{7}{2}$$

Therefore, $\mathbf{r}(t) = \left\langle \frac{1}{2}e^{2t} - \frac{1}{2}, \ln|t+1| + 2, \frac{1}{2}e^{t^2} - \frac{7}{2} \right\rangle$.

11. Let $\mathbf{r}'(t) = \frac{1}{1+t^2}\mathbf{i} - \frac{1}{\sqrt{1-t^2}}\mathbf{j} + \frac{t}{t^2+1}\mathbf{k}$ and $\mathbf{r}(0) = \mathbf{i} + \frac{\pi}{2}\mathbf{j} + 2\mathbf{k}$. Determine $\mathbf{r}(t)$.

We need to compute $\int \mathbf{r}'(t) dt$. We calculate:

$$\begin{aligned}\mathbf{r}(t) &= \int \mathbf{r}'(t) dt \\ &= \left\langle \int \frac{1}{1+t^2} dt, -\int \frac{1}{\sqrt{1-t^2}} dt, \int \frac{t}{t^2+1} dt \right\rangle \\ &= \left\langle \tan^{-1}(t) + C_1, -\sin^{-1}(t) + C_2, \frac{1}{2} \ln|t^2+1| + C_3 \right\rangle\end{aligned}$$

Since $\mathbf{r}(0) = \left\langle 1, \frac{\pi}{2}, 2 \right\rangle$ this implies that

$$0 + C_1 = 1 \quad \rightarrow \quad C_1 = 1$$

$$0 + C_2 = \frac{\pi}{2} \quad \rightarrow \quad C_2 = \frac{\pi}{2}$$

$$0 + C_3 = 2 \quad \rightarrow \quad C_3 = 2$$

Therefore, $\mathbf{r}(t) = \left\langle \tan^{-1}(t) + 1, \frac{\pi}{2} - \sin^{-1}(t), \frac{1}{2} \ln|t^2+1| + 2 \right\rangle$.

12. Suppose that we have vector functions $\mathbf{u}$ and $\mathbf{v}$ where $\mathbf{u}(2) = \langle 1, 2, -1 \rangle$, $\mathbf{u}'(2) = \langle 3, 0, 4 \rangle$, and $\mathbf{v}(t) = \langle t, t^2, t^3 \rangle$.

a) Find $f'(2)$, where $f(t) = \mathbf{u}(t) \cdot \mathbf{v}(t)$.

We calculate that $f'(t) = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$.

Therefore,

$$\begin{aligned}f'(2) &= \mathbf{u}'(2) \cdot \mathbf{v}(2) + \mathbf{u}(2) \cdot \mathbf{v}'(2) = \langle 3, 0, 4 \rangle \cdot \langle 2, 4, 8 \rangle + \langle 1, 2, -1 \rangle \cdot \langle 1, 4, 12 \rangle \\ &= (6 + 0 + 32) + (1 + 8 - 12) = (38) + (-3) = 35\end{aligned}$$

b) Find $\mathbf{r}'(2)$, where $\mathbf{r}(t) = \mathbf{u}(t) \times \mathbf{v}(t)$.

We calculate that $\mathbf{r}'(t) = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$.

Therefore,

$$\begin{aligned}\mathbf{r}'(2) &= \mathbf{u}'(2) \times \mathbf{v}(2) + \mathbf{u}(2) \times \mathbf{v}'(2) = \langle 3, 0, 4 \rangle \times \langle 2, 4, 8 \rangle + \langle 1, 2, -1 \rangle \times \langle 1, 4, 12 \rangle \\ &= \langle -16, -16, 12 \rangle + \langle 28, -13, 2 \rangle = \langle 12, -29, 14 \rangle\end{aligned}$$

13. Show that if $\mathbf{r}$ is a vector function such that $\mathbf{r}''$ exists, then $\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{r}'(t)] = \mathbf{r}(t) \times \mathbf{r}''(t)$.

We calculate that

$$\begin{aligned}\frac{d}{dt}[\mathbf{r}(t) \times \mathbf{r}'(t)] &= \frac{d}{dt}[\mathbf{r}(t)] \times \mathbf{r}'(t) + \mathbf{r}(t) \times \frac{d}{dt}[\mathbf{r}'(t)] \\ &= [\mathbf{r}'(t) \times \mathbf{r}'(t)] + [\mathbf{r}(t) \times \mathbf{r}''(t)] \\ &= \mathbf{0} + [\mathbf{r}(t) \times \mathbf{r}''(t)] \\ &= \mathbf{r}(t) \times \mathbf{r}''(t)\end{aligned}$$

14. Find an expression for $\frac{d}{dt}[\mathbf{u}(t) \cdot (\mathbf{v}(t) \times \mathbf{w}(t))]$.

$$\begin{aligned}\frac{d}{dt}[\mathbf{u}(t) \cdot (\mathbf{v}(t) \times \mathbf{w}(t))] &= \frac{d}{dt}[\mathbf{u}(t)] \cdot (\mathbf{v}(t) \times \mathbf{w}(t)) + \mathbf{u}(t) \cdot \frac{d}{dt}[\mathbf{v}(t) \times \mathbf{w}(t)] \\ &= \mathbf{u}'(t) \cdot (\mathbf{v}(t) \times \mathbf{w}(t)) + \mathbf{u}(t) \cdot [(\mathbf{v}'(t) \times \mathbf{w}(t)) + (\mathbf{v}(t) \times \mathbf{w}'(t))] \\ &= \mathbf{u}'(t) \cdot (\mathbf{v}(t) \times \mathbf{w}(t)) + \mathbf{u}(t) \cdot (\mathbf{v}'(t) \times \mathbf{w}(t)) + \mathbf{u}(t) \cdot (\mathbf{v}(t) \times \mathbf{w}'(t))\end{aligned}$$

Note: This pattern mirrors exactly what happens with we take the derivative of the product of 3 scalar function $\frac{d}{dx}[f(x)g(x)h(x)] = f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x)$. Note that each of the three terms has exactly one “prime” on it.